

Richardson's Constants

Here, we recall Richardson's constants obtained in his work [74]. See section 1.4.3 for explanations of what are all these constants and parameters.

$$\begin{aligned}a_{21} &= \frac{3c_3(\kappa^2 - 2)}{4(1 + 2c_2)}, \\a_{22} &= \frac{3c_3}{4(1 + 2c_2)}, \\a_{23} &= -\frac{3c_3\lambda}{4\kappa d_1} [3\kappa^3\lambda - 6\kappa(\kappa - \lambda) + 4], \\a_{24} &= -\frac{3c_3\lambda}{4\kappa d_1} (2 + 3\kappa\lambda) \\a_{31} &= -\frac{9\lambda}{4d_2} (4c_3(\kappa a_{23} - b_{21}) + \kappa c_4(4 + \kappa^2)) + \left(\frac{9\lambda^2 + 1 - c_2}{2d_2} \right) (3c_3(2a_{23} - \kappa b_{21}) + c_4(2 + 3\kappa^2)), \\a_{32} &= -\frac{1}{d^2} \left(\frac{9\lambda}{4} (4c_3(\kappa a_{24} - b_{22}) + \kappa c_4) + \frac{3}{2} (9\lambda^2 + 1 - c_2) (c_3(\kappa b_{22} + d_{21} - 2a_{24}) - c_4) \right), \\b_{21} &= -\frac{3c_3\lambda}{2d_1} (3\kappa\lambda - 4), \\b_{22} &= \frac{3c_3\lambda}{d_1}, \\b_{31} &= \frac{3}{8d_2} \left(8\lambda (3c_3(\kappa b_{21} - 2a_{23}) - c_4(2 + 3\kappa^2)) + (9\lambda^2 + 1 + 2c_2) (4c_3(\kappa a_{23} - b_{21}) + \kappa c_4(4 + \kappa^2)) \right), \\b_{32} &= \frac{1}{d_2} \left(9\lambda (3c_3(\kappa b_{22} + d_{21} - 2a_{24}) - c_4) + \frac{3}{8} (9\lambda^2 + 1 + 2c_2) (4c_3(\kappa a_{24} - b_{22}) + \kappa c_4) \right) \\d_{21} &= -\frac{c_3}{2\lambda^2}, \\d_{31} &= \frac{3}{64\lambda^2} (4c_3 a_{24} + c_4), \\d_{32} &= \frac{3}{64\lambda^2} (4c_3 a_{23} - d_{21} + c_4(4 + \kappa^2)),\end{aligned}$$

where

$$d_1 = \frac{3\lambda^2}{\kappa} (\kappa(6\lambda^2 - 1) - 2\lambda)$$

and

$$d_2 = \frac{8\lambda^2}{\kappa} (\kappa(11\lambda^2 - 1) - 2\lambda)$$

s_1 and s_2 are defined by

$$s_1 = \frac{\frac{3}{2}c_3(2a_{21}(\kappa^2 - 2) - a_{23}(\kappa^2 + 2) - 2\kappa b_{21}) - \frac{3}{8}(3\kappa^4 - 8\kappa^2 + 8)}{2\lambda(\lambda(1 + \kappa^2) - 2\kappa)},$$

$$s_2 = \frac{\frac{3}{2}c_3(2a_{22}(\kappa^2 - 2) - a_{24}(\kappa^2 + 2) + 2\kappa b_{22} + 5d_{21}) + \frac{3}{8}c_4(12 - \kappa^2)}{2\lambda(\lambda(1 + \kappa^2) - 2\kappa)},$$

where $\kappa = \frac{1}{2\lambda}(\lambda^2 + 1 + 2c_2)$, and λ is solution of

$$\lambda^4 + (c_2 - 2)\lambda^2 - (c_2 - 1)(1 + 2c_2) = 0.$$

Constants l_1 and l_2 are defined by

$$l_1 = a_1 + 2l^2s_1 \quad \text{et} \quad l_2 = a_2 + 2l^2s_2,$$

where $a_1 = -\frac{3}{2}c_3(2a_{21} + a_{23} + 5d_{21}) - \frac{3}{8}c_4(12 - k^2)$ and $a_2 = \frac{3}{2}(a_{24} - 2a_{22}) + \frac{9}{8}c_4$.